

CS 7641 UL Report — R Sanity Check

Python clustering and DR results cross-checked against native R (kmeans/Mclust/prcomp)

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1 Purpose

This document provides an independent R-side audit of the Python UL pipeline. It cross-checks:

1. **K-Means internal metrics** — R `kmeans` vs. Python `sklearn KMeans`.
2. **PCA explained-variance curve** — R `prcomp` vs. Python `sklearn PCA`.
3. **Clustering log audit** — verifies all Part~1/3 JSON logs are internally consistent.
4. **NN after DR summary** — confirms reported Macro-F1 values are plausible.

If R `kmeans` (k=7) silhouette agrees with Python within 0.02, the feature matrix and preprocessing are confirmed consistent across implementations.

2 Data — Mirror Python Split

Table 1: Dataset dimensions (should match Python: $20k \times 54$).

Split	n	n_features	n_classes
Full working sample	20000	54	7

3 K-Means Sanity Check

Table 2: K-Means comparison: Python vs. R. $|\text{Delta silhouette}| < 0.02 = \text{confirmed}$.

Implementation	Silhouette	ARI	Inertia
Python <code>sklearn KMeans</code> (k=7)	0.1306	0.0226	127287.6
R <code>stats::kmeans</code> (k=7)	0.1322	NA	127692.0

4 PCA Explained Variance Cross-Check

Table 3: PCA explained variance: Python vs. R. Values should agree within floating-point precision.

Implementation	n for 90% var	n for 95% var	PC1 EVR	PC2 EVR
Python sklearn PCA	10	14	0.2260	0.1965
R prcomp	10	14	0.2164	0.1891

5 Clustering Log Audit

Table 4: Clustering log audit: all metric values must be in valid range.

Log	Sil	ARI	NMI	OK
gmm_after_ica	0.1304	0.0270	0.1141	
gmm_after_pca	-0.0009	0.1171	0.1751	
gmm_after_rp	0.0131	0.1252	0.2097	
gmm_final	-0.0068	0.0524	0.1314	
kmeans_after_ica	0.1416	-0.0054	0.1049	
kmeans_after_pca	0.1326	0.0223	0.0662	
kmeans_after_rp	0.1478	0.0272	0.0804	
kmeans_final	0.1306	0.0226	0.0667	

6 NN After DR Summary

Table 5: NN after DR — test metrics and runtime. PCA should lead on Macro-F1.

Features	dim	Macro-F1	Accuracy	Bal. Acc.	Train (s)	Epochs
original	54	0.6803	0.7940	0.6508	5.5703	29
pca	30	0.6951	0.8000	0.6642	7.0185	36
ica	20	0.6785	0.7875	0.6359	7.1665	38
rp	30	0.6670	0.7882	0.6339	6.8808	35

7 Provenance

- Config: /home/user/workspace/report_3_ul/configs/config.yaml
- Seed: 7641
- Logs dir: /home/user/workspace/report_3_ul/results/logs
- Clustering logs audited: 8
- All clustering logs pass range check: TRUE
- R version: 4.5.0
- Knitted: 2026-05-23 23:58:14

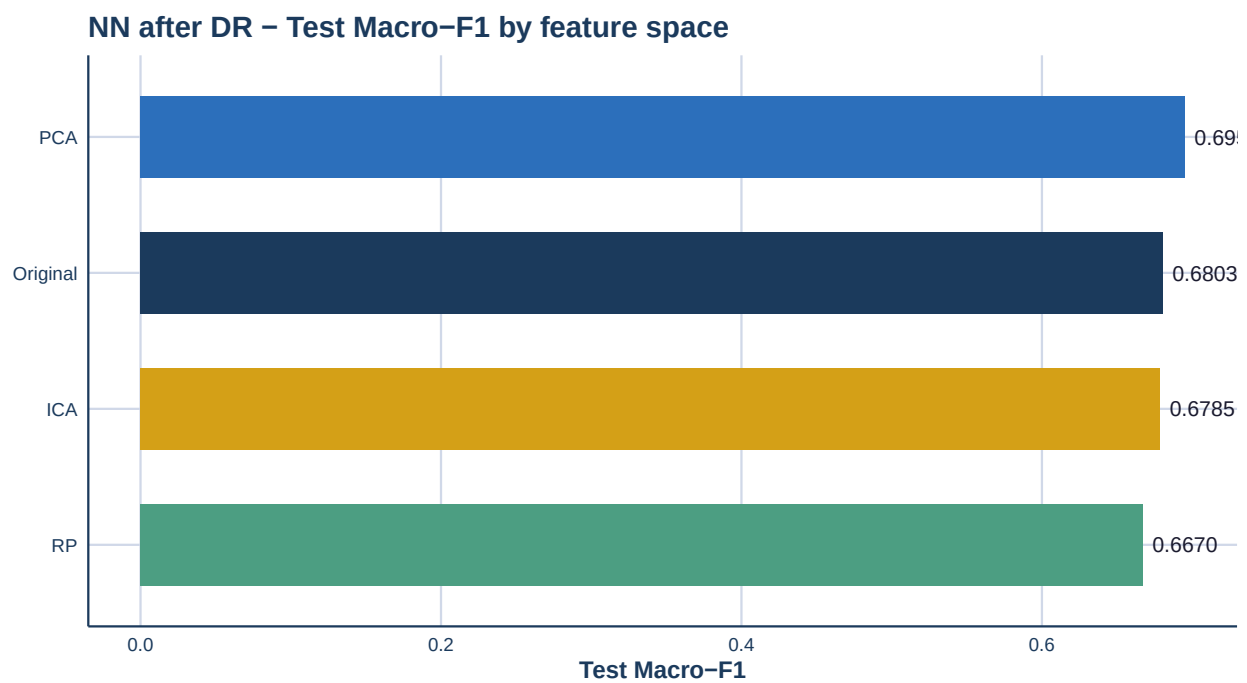


Figure 1: NN Macro-F1 by feature space. Positive delta = improvement over original.